

Lecture 6: The Data Table as a Matrix

One table of numbers, ten things we can do to it — and what each one says about the countries

STAT 481 — Multivariate Analysis

Sun, Sep 13, 2026 • 50 minutes

Today's Overview

- 1. The data today.** We read the file `data/whwide2022.csv` and keep **four of its columns**: `pop0014` (share of the population under 15, %), `pop65` (share aged 65 and over, %), `life_exp` (life expectancy, years) and `gdp_pc` (income per person, used in thousand US\$). **All 175 rows** — one per country, named by `country`. Two percentages, a number of years, an amount of money: mixed units on purpose.
- 2. The whole table is one object.** We stop treating the file as 175 rows or 4 columns and treat it as *one* thing that can be flipped, multiplied and combined.
- 3. Three black boxes opened.** `colMeans()`, `cov()` and `cor()` have been called since Week 1; today each one is written as a single product of the data table with itself, and checked against R to the last digit.
- 4. Four numbers that describe a covariance table.** How many of its columns carry new information; how much the whole table varies; how much it varies once the overlap between columns is counted; and the table that undoes it — with the moment R refuses to compute the last one.
- 5. The distance from Lecture 5, written properly.** The one line of R that Lecture 5 used without explaining is explained.
- 6. A rotation that changes every number and nothing that matters.** The door to Lecture 7.
- 7. Then:** a warm-up, home problems on other columns of the wide file, and an FAQ.

Learning Objectives

By the end of this lecture you should be able to:

- multiply matrices in R (`%*%`, `t()`, `crossprod()`), read the dimensions of a product before computing it, and say why `A %*% A` fails for a 2×3 matrix;
- derive the covariance matrix as one product of the centred data matrix, $\mathbf{S} = \mathbf{X}_c^\top \mathbf{X}_c / (n - 1)$, and the correlation matrix as a diagonal sandwich of it — and check both against R;
- explain why the correlation between two variables is the cosine of an angle, and why that forces $-1 \leq r \leq 1$;
- compute and interpret $\text{rank}(\mathbf{S})$, $\text{tr}(\mathbf{S})$, $|\mathbf{S}|$, $|\mathbf{R}|$ and $\mathbf{S}^{-1}$ for a data set, and predict which of them breaks when a column is a sum of two others;
- state what an orthogonal matrix does to a data cloud and which summaries it leaves untouched.

Packages today

- **Base R only** [known]: `as.matrix()`, `t()`, `%*%`, `crossprod()`, `diag()`, `qr()`, `det()`, `solve()`, `zapsmall()`. Nothing to install.

Open the RStudio project `multi` first, so that `"data/whwide2022.csv"` resolves. Indicators: World Bank WDI, 2022.

Notation

- **Bold capital X, S, R, H:** a matrix (a table of numbers). **Bold lower case x, 1:** a column vector. Plain x_{ij} , s_{jk} , n , p : single numbers.
- $\mathbf{X}^\top$ is the **transpose** of $\mathbf{X}$ (rows and columns swapped). J&W write it $\mathbf{X}'$; R writes it $\mathbf{t}(\mathbf{X})$.
- Under a product we write its dimensions, as J&W do: $\mathbf{X}_{(n \times p)}$.
- $n = 175$ countries, $p = 4$ variables, unless a chunk says otherwise (Section 9 revisits Lecture 5's eight columns).

1 A data set is a matrix; a variable is a column

In this section

The question. We have looked at this file as 175 rows (countries) and as 4 columns (variables). Is there a way to hold the whole thing at once?

What we do about it. Call the table a *matrix*, give its rows and columns names, learn one move — the transpose — and use it to write the mean vector as a single product.

1. Definition

Definition 1.1: Matrix, column vector, transpose, the vector 1, the identity

- An $n \times p$ **matrix** is a rectangular array of n rows and p columns. Entry x_{ij} sits in row i , column j .

$$\mathbf{X}_{(n \times p)} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}$$

- A matrix with one column is a **column vector**; with one row, a **row vector**. Column j of $\mathbf{X}$ is $\mathbf{x}_j$ ($n \times 1$); row i is $\mathbf{x}_i^\top$ ($1 \times p$).
- The **transpose** $\mathbf{X}^\top$ turns every column into a row: it is $p \times n$, with entry (j, i) equal to x_{ij} .

$$(\mathbf{X}^\top)^\top = \mathbf{X}, \quad (\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$$

- $\mathbf{1}$ is the $n \times 1$ column of ones. $\mathbf{I}$ is the square matrix with ones on the diagonal and zeros elsewhere; $\mathbf{IA} = \mathbf{AI} = \mathbf{A}$ for any $\mathbf{A}$.
- **Tiny example** (J&W Example 2.3):

$$\mathbf{A} = \begin{pmatrix} 3 & -1 & 2 \\ 1 & 5 & 4 \end{pmatrix} \text{ is } 2 \times 3, \quad \mathbf{A}^\top = \begin{pmatrix} 3 & 1 \\ -1 & 5 \\ 2 & 4 \end{pmatrix} \text{ is } 3 \times 2.$$

2. In statistics

Fact 1.1: The data array is $\mathbf{X}$, and the mean vector is a product

- Lecture 1's data array is $\mathbf{X}$: one row per country, one column per variable. $\mathbf{x}_j$ holds all 175 values of variable j ; $\mathbf{x}_i^\top$ holds the four values of country i .
- Adding the entries of a column is a product with $\mathbf{1}$: $\mathbf{x}_j^\top \mathbf{1} = \sum_i x_{ij}$. Doing it for all columns at once:

$$\bar{\mathbf{x}} = \frac{1}{n} \mathbf{X}_{(p \times n)}^\top \mathbf{1}_{(n \times 1)} \quad \text{— the } p \times 1 \text{ vector of column means.}$$

3. In R

Today's four columns, with income in **thousand** dollars so that every table in this lecture fits on the page (Section 4 puts the dollars back):

```
options(digits = 4)
w <- read.csv("data/whwide2022.csv", stringsAsFactors = FALSE) # L5
X <- as.matrix(w[, c("pop0014", "pop65", "life_exp", "gdp_pc")])
X[, "gdp_pc"] <- X[, "gdp_pc"] / 1000 # income in THOUSAND US$
colnames(X)[4] <- "gdp_k"
rownames(X) <- w$country
dim(X)
round(X[c("Qatar", "Japan", "Niger"), ], 2)
```

```
[1] 175 4
      pop0014 pop65 life_exp gdp_k
Qatar   15.25  1.39   81.86 88.70
Japan   11.86 29.40   84.00 35.55
Niger   47.33  2.55   60.40  0.61
```

- `as.matrix(w[, cols])` takes the named columns of the data frame and returns a plain numeric matrix — from here on the object has no “columns of a data frame”, only entries x_{ij} .
- Read the three printed rows as three $\mathbf{x}_i^\top$: Qatar is young and rich, Japan old and rich, Niger very young and poor.

```
n <- nrow(X); p <- ncol(X); c(n = n, p = p)
x1 <- X[, 1] # first COLUMN: one variable, 175 numbers
length(x1)
X["Qatar", ] # one ROW: one country, four numbers
```

```
  n  p
175 4
[1] 175
  pop0014  pop65 life_exp  gdp_k
  15.248   1.391  81.857  88.701
```

- `nrow()`, `ncol()` return n and p ; `X[, 1]` is the column vector $\mathbf{x}_1$; `X["Qatar", ]` is the row $\mathbf{x}_i^\top$ for Qatar.

```
A <- matrix(c(3, 1, -1, 5, 2, 4), nrow = 2) # J&W Example 2.3, a 2 x 3
A
t(A) # rows become columns: a 3 x 2
dim(A); dim(t(A))
```

```
      [,1] [,2] [,3]
[1,]    3   -1    2
[2,]    1    5    4
      [,1] [,2]
[1,]    3    1
[2,]   -1    5
[3,]    2    4
[1] 2 3
[1] 3 2
```

- `matrix(v, nrow = 2)` fills the six numbers **column by column** — so the first two numbers form column 1. `t()` swaps rows and columns; `dim()` reports (rows, columns), and the transpose reverses the pair.

Now the mean vector as a product — `one` is a vector of 175 ones, so the result must be 4×1 :

```
one <- rep(1, n)           # the vector 1: n ones
xbar <- t(X) %*% one / n    # (1/n) X'1 -- a p x 1 column
t(xbar)                    # printed as a row to save space
colMeans(X)                # the shortcut
all.equal(as.vector(xbar), unname(colMeans(X)))
```

```
      pop0014 pop65 life_exp gdp_k
[1,]      26.6  9.741    72.53 18.45
      pop0014      pop65 life_exp      gdp_k
      26.599      9.741    72.526    18.451
[1] TRUE
```

- `rep(1, n)` builds `1`; `%*%` is the matrix product; the result is $p \times 1$, printed here through `t()` as one row.
- The four numbers are exactly `colMeans(X)`: on average 26.6% of a country's people are under 15, 9.7% over 65, life expectancy is 72.5 years and income is \$18 451 per person.
- `all.equal()` compares two numeric objects up to rounding and says `TRUE` or reports the difference — today's checking tool, used in every section.

2 Products and conformability

In this section

The question. *What does it mean to multiply two tables — and when is it even allowed?*

What we do about it. The row-by-column rule, the dimension bookkeeping that predicts whether a product exists and how big it is, and the two products of vectors that the rest of the course lives on. Then the error message R gives when the bookkeeping fails.

1. Definition

Definition 2.1: Matrix product, inner product, outer product

- $\mathbf{AB}$ exists only when the **inner** dimensions match: $\mathbf{A}$ is $n \times k$ and $\mathbf{B}$ is $k \times p$. The result is $n \times p$ (the **outer** dimensions).
- Entry (i, j) is row i of $\mathbf{A}$ against column j of $\mathbf{B}$ — multiply pairwise and add (J&W 2-10):

$$(\mathbf{AB})_{ij} = \sum_{\ell=1}^k a_{i\ell} b_{\ell j}, \quad \underset{(n \times k)}{\mathbf{A}} \underset{(k \times p)}{\mathbf{B}} = \underset{(n \times p)}{\mathbf{AB}}.$$

- Order matters: $\mathbf{AB}$ and $\mathbf{BA}$ are different objects, and often only one of them exists.
- For two column vectors $\mathbf{x}, \mathbf{y}$ of length n :

$$\underset{(1 \times n)}{\mathbf{x}^\top} \underset{(n \times 1)}{\mathbf{y}} = \sum_{i=1}^n x_i y_i \quad (\text{inner product: one number})$$

$$\underset{(n \times 1)}{\mathbf{x}} \underset{(1 \times n)}{\mathbf{y}^\top} \quad (\text{outer product: an } n \times n \text{ table}).$$

- **Tiny example.** With $\mathbf{A}$ (2×3) from Section 1 and $\mathbf{B} = \begin{pmatrix} 1 & -1 \\ 0 & 3 \\ 2 & 1 \end{pmatrix}$ (3×2):

$$\mathbf{AB} = \begin{pmatrix} 3 \cdot 1 + (-1) \cdot 0 + 2 \cdot 2 & 3 \cdot (-1) + (-1) \cdot 3 + 2 \cdot 1 \\ 1 \cdot 1 + 5 \cdot 0 + 4 \cdot 2 & 1 \cdot (-1) + 5 \cdot 3 + 4 \cdot 1 \end{pmatrix} = \begin{pmatrix} 7 & -4 \\ 9 & 18 \end{pmatrix}.$$

2. In statistics**Fact 2.1: Sums of squares and cross-products are inner products**

- $\mathbf{x}_j^\top \mathbf{x}_j = \sum_i x_{ij}^2$ is the **sum of squares** of variable j ; $\mathbf{x}_j^\top \mathbf{x}_k = \sum_i x_{ij} x_{ik}$ is the **cross-product** of variables j and k .
- $\mathbf{X}^\top \mathbf{X}$ is the $p \times p$ table of *all* sums of squares (diagonal) and cross-products (off-diagonal) at once. Section 3 turns it into $\mathbf{S}$.

3. In R

```
B <- matrix(c(1, 0, 2, -1, 3, 1), nrow = 3) # a 3 x 2
A %*% B # (2 x 3)(3 x 2) = 2 x 2
B %*% A # (3 x 2)(2 x 3) = 3 x 3: different
all.equal(t(A %*% B), t(B) %*% t(A)) # (AB)' = B'A'
```

```
      [,1] [,2]
[1,]    7  -4
[2,]    9  18
      [,1] [,2] [,3]
[1,]    2  -6  -2
[2,]    3  15  12
[3,]    7   3   8
[1] TRUE
```

- The 2×2 is the hand calculation above, entry for entry. $\mathbf{B} \%*\% \mathbf{A}$ also exists (3×2 times 2×3) and is a 3×3 : a different matrix, not a reordering of the first.
- The last line checks $(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$ — the transpose of a product reverses the order.

```
u <- X[, "pop65"]; v <- X[, "life_exp"]      # two columns, 175 each
t(u) %*% v                                # inner product u'v: ONE number (a 1 x 1)
sum(u * v)                                # the same number, written as a sum
crossprod(u, v)                            # the same again: R's name for t(u) %*% v
dim(u %*% t(v))                            # outer product uv': a 175 x 175 table
```

```
      [,1]
[1,] 130365
[1] 130365
      [,1]
[1,] 130365
[1] 175 175
```

- Three ways to write one number: $t(u) \%*\% v$ (a 1×1 matrix), $sum(u * v)$ (the definition), $crossprod(u, v)$ — R's name for $u^T v$, one argument or two.
- The outer product $u \%*\% t(v)$ of the same two columns is a 175×175 table — same ingredients, opposite shape. Which one you get is decided by the dimensions, not by the data.

```
round(crossprod(X))                        # X'X: every column against every column
```

	pop0014	pop65	life_exp	gdp_k
pop0014	142992	34675	325318	57919
pop65	34675	25485	130365	52207
life_exp	325318	130365	932209	256841
gdp_k	57919	52207	256841	199061

- $crossprod(X)$ with one argument is $t(X) \%*\% X$: the 4×4 table of raw sums of squares and cross-products. The `pop65-life_exp` entry is the 130365 of the previous chunk.
- Raw sums grow with n and carry the units squared — not yet a summary anyone reads. Two sections from now it becomes **S**.

? Predict 1 — answer before you read on

A is 2×3 . What does $A \%*\% A$ return?

```
A %*% A                                # (2 x 3)(2 x 3): inner dimensions 3 and 2
```

```
Error in A %*% A : non-conformable arguments
```

- Read it aloud: *non-conformable arguments* — the inner dimensions are 3 and 2, so no entry can be formed. R checks the dimensions before it looks at a single number.
- Every product in this course can be checked the same way before it is typed: write the dimensions underneath and see whether the inner pair matches.

△! * is not %*%

$A * B$ multiplies entry by entry and needs two tables of the *same* shape; $A \%*\% B$ is the matrix product of this section. For the 2×3 matrix A, $A * A$ runs and squares every entry; $A \%*\% A$ stops with the error above. They are never interchangeable.

3 Centring is a matrix product

In this section

The question. Until now $s_{jk} = \frac{1}{n-1} \sum_i (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)$ was one formula per pair of columns. Can the whole covariance table be written as one product?

What we do about it. Build the matrix that subtracts every column's mean, apply it, and multiply the result by its own transpose. **What the section leaves behind:** the identity $\mathbf{S} = \mathbf{X}_c^\top \mathbf{X}_c / (n-1)$, which every later section reuses.

1. Definition

Definition 3.1: The centring matrix and the centred data

- The **centring matrix** is the $n \times n$ matrix

$$\mathbf{H} = \mathbf{I} - \frac{1}{n} \mathbf{1}\mathbf{1}^\top.$$

- Applied to a column, $\mathbf{H}\mathbf{x}$ subtracts the mean of $\mathbf{x}$ from every entry: $\mathbf{1}\mathbf{1}^\top \mathbf{x} / n$ is the column whose every entry is $\bar{x}$.
- Two properties we use: $\mathbf{H}$ is symmetric, and centring twice is centring once.

$$\mathbf{H}^\top = \mathbf{H}, \quad \mathbf{H}\mathbf{H} = \mathbf{H}.$$

- Tiny example**, $n = 3$: for $\mathbf{x} = (1, 2, 6)^\top$, the mean is 3, so

$$\mathbf{H} = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \end{pmatrix}, \quad \mathbf{H}\mathbf{x} = \begin{pmatrix} -2 \\ -1 \\ 3 \end{pmatrix}.$$

2. In statistics

Fact 3.1: S is one product of the centred data

- $\mathbf{X}_c = \mathbf{H}\mathbf{X}$ is the **centred data matrix**: every column has had its own mean removed, so every column of $\mathbf{X}_c$ sums to 0.
- The sample covariance matrix is

$$\mathbf{S} = \frac{1}{n-1} \mathbf{X}^\top \mathbf{H}\mathbf{X} = \frac{1}{n-1} \underset{(p \times n)}{\mathbf{X}_c^\top} \underset{(n \times p)}{\mathbf{X}_c} \quad (\text{J\&W 3-27})$$

(the middle form uses $\mathbf{H}^\top \mathbf{H} = \mathbf{H}\mathbf{H} = \mathbf{H}$).

- Entry (j, k) of $\mathbf{X}_c^\top \mathbf{X}_c$ is the inner product of two centred columns, $\sum_i (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)$ — the formula for s_{jk} , all p^2 of them in one product.

3. In R

```
H3 <- diag(3) - matrix(1, 3, 3) / 3      # I - (1/n) 11' for n = 3
H3
H3 %*% c(1, 2, 6)                        # subtracts the mean 3 from every entry
```

```
      [,1] [,2] [,3]
[1,]  0.6667 -0.3333 -0.3333
[2,] -0.3333  0.6667 -0.3333
[3,] -0.3333 -0.3333  0.6667
      [,1]
[1,]    -2
[2,]    -1
[3,]     3
```

- `diag(3)` is $\mathbf{I}_3$; `matrix(1, 3, 3)` is $\mathbf{1}\mathbf{1}^\top$ (a 3×3 of ones). The printed $\mathbf{H}$ is the one in the definition, and `H3 %*% c(1, 2, 6)` returns $(-2, -1, 3)$: the mean 3 removed from each entry.

? Predict 2 — answer before you read on

`S_hand <- t(Xc) %*% Xc / (n - 1)` on the 175 centred rows. Will `all.equal(S_hand, cov(X))` say TRUE — and what will the entry for `pop0014` against `pop65` look like: positive or negative?

```
H <- diag(n) - matrix(1, n, n) / n      # 175 x 175 centring matrix
Xc <- H %*% X                            # every column minus its own mean
round(colMeans(Xc), 10)                  # centred: every column mean is 0
S_hand <- t(Xc) %*% Xc / (n - 1)         # S = Xc'Xc / (n - 1)
round(S_hand, 3)
all.equal(S_hand, cov(X))
```

```
pop0014    pop65 life_exp    gdp_k
      0         0         0         0
      pop0014    pop65 life_exp    gdp_k
pop0014    110.22 -61.31   -70.56 -160.7
pop65      -61.31  51.03    38.68  119.3
life_exp   -70.56  38.68    67.28  130.2
gdp_k      -160.73 119.28   130.24  801.6
[1] TRUE
```

- `diag(n)` is the 175×175 identity; $\mathbf{H}$ has $175^2 = 30\,625$ entries; `H %*% X` centres all four columns in one product, and the column means of `Xc` print as 0 (`round(, 10)` hides rounding dust of order 10^{-15}).
- **TRUE: the formula and `cov()` agree to the last digit.** From now on `cov(X)` is not a black box; it is $\mathbf{X}_c^\top \mathbf{X}_c / 174$.
- Read $\mathbf{S}$ itself. The diagonal holds the four variances: income (thousand dollars, squared) has 801.6, the largest by far; the share of over-65s has 51.0. Off the diagonal, `pop0014–pop65` is -61.3 : countries with many children have few over-65s.

```
all.equal(H %*% H, H)                    # centring twice = centring once
all.equal(Xc, scale(X, center = TRUE, scale = FALSE),
          check.attributes = FALSE)      # the shortcut
```

```
[1] TRUE
[1] TRUE
```

- `H %*% H` is `H`: centring already-centred data changes nothing. (The same fact makes $\mathbf{X}^\top \mathbf{H} \mathbf{X}$ equal $\mathbf{X}_c^\top \mathbf{X}_c$.)
- `scale(X, center = TRUE, scale = FALSE)` is the shortcut: `center = TRUE` subtracts the column means, `scale = FALSE` leaves the spreads alone. `check.attributes = FALSE` tells `all.equal()` to ignore the extra labels `scale()` attaches and compare only the numbers.

4 Diagonal matrices rescale; $\mathbf{R}$ is a sandwich

In this section

The question. A correlation is a covariance “in standard-deviation units”. What is that as a matrix operation — and what happens to $\mathbf{S}$ if we put the dollars back?

What we do about it. Multiply by a diagonal matrix on the left, on the right, and on both sides; watch what each does. Then build $\mathbf{R}$ from $\mathbf{S}$ and $\mathbf{S}$ from $\mathbf{R}$.

1. Definition

Definition 4.1: Diagonal matrix; the sandwich

- A **diagonal matrix** has zeros everywhere except on its diagonal: $\mathbf{D} = \text{diag}(d_1, \dots, d_p)$.
- Multiplying from the left scales the *rows*; from the right, the *columns*; on both sides, entry (i, j) by $d_i d_j$:

$$(\mathbf{D}\mathbf{A})_{ij} = d_i a_{ij}, \quad (\mathbf{A}\mathbf{D})_{ij} = a_{ij} d_j, \quad (\mathbf{D}\mathbf{A}\mathbf{D})_{ij} = d_i a_{ij} d_j.$$

- Powers of a diagonal matrix act entry by entry: $\mathbf{D}^{1/2} = \text{diag}(\sqrt{d_j})$, $\mathbf{D}^{-1/2} = \text{diag}(1/\sqrt{d_j})$.
- **Tiny example.** With $\mathbf{D} = \text{diag}(2, 10)$ and $\mathbf{M}$ the 2×2 matrix of ones:

$$\mathbf{D}\mathbf{M} = \begin{pmatrix} 2 & 2 \\ 10 & 10 \end{pmatrix}, \quad \mathbf{M}\mathbf{D} = \begin{pmatrix} 2 & 10 \\ 2 & 10 \end{pmatrix}, \quad \mathbf{D}\mathbf{M}\mathbf{D} = \begin{pmatrix} 4 & 20 \\ 20 & 100 \end{pmatrix}.$$

2. In statistics

Fact 4.1: Correlation is covariance between two diagonal matrices

- Let $\mathbf{D} = \text{diag}(s_{11}, \dots, s_{pp})$ hold the variances (the diagonal of $\mathbf{S}$). Then $\mathbf{D}^{-1/2}$ holds $1/s_j$, and

$$\mathbf{R} = \mathbf{D}^{-1/2} \mathbf{S} \mathbf{D}^{-1/2}, \quad \mathbf{S} = \mathbf{D}^{1/2} \mathbf{R} \mathbf{D}^{1/2} \quad (\text{J\&W 3-29, 3-30}).$$

- Entry by entry this is $r_{jk} = s_{jk}/(s_j s_k)$ — the definition of a correlation, all p^2 of them in one sandwich.
- Changing the *unit* of a variable is the same operation: income in dollars instead of thousand dollars is $\mathbf{D}\mathbf{S}\mathbf{D}$ with $\mathbf{D} = \text{diag}(1, 1, 1, 1000)$. $\mathbf{S}$ changes; $\mathbf{R}$ does not.

3. In R

```
M <- matrix(1, 2, 2); D2 <- diag(c(2, 10))
D2 %**% M           # D on the LEFT: row i times d_i
M %**% D2           # D on the RIGHT: column j times d_j
D2 %**% M %**% D2   # both sides: entry (i, j) times d_i * d_j
```

```
      [,1] [,2]
[1,]     2     2
[2,]    10    10
      [,1] [,2]
[1,]     2    10
[2,]     2    10
      [,1] [,2]
[1,]     4    20
[2,]    20   100
```

- `diag(c(2, 10))` builds the diagonal matrix from a vector; with a single number, `diag(3)`, it builds the identity; with a matrix, `diag(S)`, it *extracts* the diagonal. Three jobs, one name — read the argument.

```
S <- cov(X)
Dm12 <- diag(1 / sqrt(diag(S)))      # D^{-1/2}: 1/s_j on the diagonal
R_hand <- Dm12 %**% S %**% Dm12      # the sandwich
dimnames(R_hand) <- dimnames(S)
round(R_hand, 3)
all.equal(R_hand, cor(X))
D12 <- diag(sqrt(diag(S)))           # D^{1/2}: s_j on the diagonal
all.equal(D12 %**% cor(X) %**% D12, S, check.attributes = FALSE)
all.equal(cov(scale(X)), cor(X), check.attributes = FALSE)
```

```
      pop0014 pop65 life_exp gdp_k
pop0014   1.000 -0.818  -0.819 -0.541
pop65    -0.818  1.000   0.660  0.590
life_exp  -0.819  0.660   1.000  0.561
gdp_k     -0.541  0.590   0.561  1.000
[1] TRUE
[1] TRUE
[1] TRUE
```

- `1 / sqrt(diag(S))` is the vector of $1/s_j$; `diag()` of it is $\mathbf{D}^{-1/2}$. The sandwich reproduces `cor(X)` exactly (TRUE), and $\mathbf{D}^{1/2}\mathbf{R}\mathbf{D}^{1/2}$ rebuilds **S** (TRUE).
- The third TRUE is the reason `scale()` appeared in Lecture 5: standardised columns have variance 1, so their covariance matrix *is* the correlation matrix.
- Read **R**: children and over-65s, -0.818 ; children and life expectancy, -0.819 ; income and everything else, moderate.

? Predict 3 — answer before you read on

Put income back in dollars (multiply the last column by 1000). Which entries of **S** change, by what factor — and does **R** change?

```
Dd <- diag(c(1, 1, 1, 1000)) # income back to dollars, rest untouched
round((Dd %*% S %*% Dd)[4, ]) # the income row of S, in dollars
round(cov(X %*% Dd)[4, ])      # the same row, from the rescaled data
all.equal(cor(X %*% Dd), cor(X), check.attributes = FALSE) # R unmoved
```

```
[1] -160728 119277 130238 801635374
[1] -160728 119277 130238 801635374
[1] TRUE
```

- The three covariances with income are multiplied by 1000; the income variance by $1000^2 = 801.6$ becomes 801 635 374. `Dd %*% S %*% Dd` predicted the row that `cov(X %*% Dd)` computed.
- `cor()` returns `TRUE`: the sandwich divides the factor back out. This is why every eight-column table in Lecture 5 was read through `R` or through `scale()`, never through raw `S`.

5 Length, inner product, angle

In this section

The question. *Lecture 5 promised that $|r| \leq 1$ is the Cauchy–Schwarz inequality. What does that mean — what is r , geometrically?*

What we do about it. Give a column of 175 numbers a length and give two columns an angle. The correlation turns out to be the cosine of that angle, and a cosine cannot leave $[-1, 1]$.

1. Definition

Definition 5.1: Length, unit vector, angle, projection (J&W 2-1 to 2-9)

- The **length** of $\mathbf{x}$ ($n \times 1$) is

$$L_{\mathbf{x}} = \sqrt{\mathbf{x}^T \mathbf{x}} = \sqrt{x_1^2 + \cdots + x_n^2},$$

and $\mathbf{x}/L_{\mathbf{x}}$ is the **unit vector** in the direction of $\mathbf{x}$ (length 1).

- The **angle** θ between $\mathbf{x}$ and $\mathbf{y}$ is defined through its cosine:

$$\cos \theta = \frac{\mathbf{x}^T \mathbf{y}}{L_{\mathbf{x}} L_{\mathbf{y}}} = \frac{\mathbf{x}^T \mathbf{y}}{\sqrt{\mathbf{x}^T \mathbf{x}} \sqrt{\mathbf{y}^T \mathbf{y}}}.$$

$\mathbf{x}$ and $\mathbf{y}$ are **perpendicular** when $\mathbf{x}^T \mathbf{y} = 0$.

- The **projection** of $\mathbf{x}$ on $\mathbf{y}$ is the shadow of $\mathbf{x}$ along $\mathbf{y}$; it has length $L_{\mathbf{x}} |\cos \theta|$:

$$\text{proj}_{\mathbf{y}} \mathbf{x} = \frac{\mathbf{x}^T \mathbf{y}}{\mathbf{y}^T \mathbf{y}} \mathbf{y}.$$

- **Tiny example** (J&W Example 2.1): $\mathbf{x} = (1, 3, 2)^T$, $\mathbf{y} = (-2, 1, -1)^T$. Then $\mathbf{x}^T \mathbf{x} = 14$, $\mathbf{y}^T \mathbf{y} = 6$, $\mathbf{x}^T \mathbf{y} = -1$, so

$$L_{\mathbf{x}} = \sqrt{14} = 3.742, \quad L_{\mathbf{y}} = \sqrt{6} = 2.449, \quad \cos \theta = \frac{-1}{3.742 \times 2.449} = -0.109, \quad \theta = 96.3^\circ.$$

2. In statistics

Fact 5.1: Correlation is a cosine

For two *centred* columns $\mathbf{a} = \mathbf{H}\mathbf{x}_j$ and $\mathbf{b} = \mathbf{H}\mathbf{x}_k$:

- the squared length is $(n - 1)$ times the variance: $\|\mathbf{a}\|^2 = \sum_i (x_{ij} - \bar{x}_j)^2 = (n - 1) s_{jj}$;
- the inner product is $(n - 1)$ times the covariance: $\mathbf{a}^\top \mathbf{b} = (n - 1) s_{jk}$;
- so the cosine of the angle between the two centred columns is

$$\cos \theta = \frac{(n - 1) s_{jk}}{\sqrt{(n - 1) s_{jj}} \sqrt{(n - 1) s_{kk}}} = \frac{s_{jk}}{s_j s_k} = r_{jk}.$$

- **Hence** $-1 \leq r \leq 1$: a cosine cannot be anything else. This is the Cauchy–Schwarz inequality $|\mathbf{a}^\top \mathbf{b}| \leq L_{\mathbf{a}} L_{\mathbf{b}}$ that Lecture 5 used to close the triangle inequality — the same fact, now with a picture. $r = 0$ is a right angle; $r = \pm 1$ is two columns on one line.

3. In R

```
x <- c(1, 3, 2); y <- c(-2, 1, -1)           # J&W Example 2.1
Lx <- sqrt(sum(x^2)); Ly <- sqrt(sum(y^2))
c(Lx = Lx, Ly = Ly, cos_theta = sum(x * y) / (Lx * Ly))
acos(sum(x * y) / (Lx * Ly)) * 180 / pi      # the angle, in degrees
```

```
      Lx      Ly cos_theta
[1] 3.7417 2.4495 -0.1091
[1] 96.26
```

- `sqrt(sum(x^2))` is $L_{\mathbf{x}}$; `acos()` returns the angle in radians, and `* 180 / pi` turns it into degrees. The book's 96.3° , to one more decimal.

? Predict 4 — answer before you read on

The angle between the centred `pop0014` column and the centred `pop65` column: more or less than 90° ? (Section 3 printed their covariance.)

```
a <- Xc[, "pop0014"]; b <- Xc[, "pop65"]      # two CENTRED columns
c(length_sq = sum(a^2), n1_var = (n - 1) * var(X[, "pop0014"]))
cos_ab <- sum(a * b) / (sqrt(sum(a^2)) * sqrt(sum(b^2)))
c(cos_theta = cos_ab, r = cor(X)[ "pop0014", "pop65" ])
acos(cos_ab) * 180 / pi                      # the angle between the two columns
```

```
length_sq    n1_var
      19179      19179
cos_theta      r
    -0.8175    -0.8175
[1] 144.8
```

- `sum(a^2)` equals $174 \times \text{Var}(\text{pop0014})$ to the last digit: the length of a centred column *is* its spread.
- `cos_theta` and `cor(X)[...]` print the same -0.8175 — the correlation is the cosine.
- 144.8° : well past a right angle. A strong negative correlation is two columns pointing in nearly opposite directions in 175-dimensional space.

6 Linear dependence and rank

In this section

The question. *Lecture 5 said that for some tables the Mahalanobis distance cannot be computed and R stops with an error. What is wrong with those tables?*

What we do about it. Define what it means for columns to be redundant, count the non-redundant ones (the rank), and build a table with a redundant column to see what it does to **S**.

1. Definition

Definition 6.1: Linear dependence; rank (J&W Definitions 2A.6, 2A.25)

- Columns $\mathbf{x}_1, \dots, \mathbf{x}_k$ are **linearly dependent** if some combination of them with coefficients not all zero is the zero vector:

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \dots + c_k\mathbf{x}_k = \mathbf{0} \quad \text{with some } c_j \neq 0.$$

Then at least one column is a combination of the others. Otherwise they are **linearly independent**.

- The **rank** of a matrix is the largest number of linearly independent columns it has — and the row count gives the same number (J&W Result 2A.6). Always

$$\text{rank}(\mathbf{A}) \leq \min(\text{rows}, \text{columns}).$$

- Tiny example.** $\begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{pmatrix}$ has rank 1: column 2 is $2 \times$ column 1, so $2\mathbf{x}_1 - \mathbf{x}_2 = \mathbf{0}$. Change the 6 to a 7 and the rank is 2.

2. In statistics

Fact 6.1: When **S** loses rank

- $\text{rank}(\mathbf{S}) = \text{rank}(\mathbf{X}_c)$, and centring costs one dimension (the columns of $\mathbf{X}_c$ all sum to zero), so

$$\text{rank}(\mathbf{S}) \leq \min(n - 1, p).$$

- A **redundant variable** — a column that is a sum, a difference or a fixed percentage of other columns — makes $\mathbf{X}_c$ and **S** lose rank. Lecture 5's wide file has several: the age bands sum to 100, urban and rural shares sum to 100.
- Too few countries** does the same: with $n \leq p$ the rank is at most $n - 1 < p$, whatever the data say.
- A $p \times p$ matrix **S** of rank less than p is called **singular**. Sections 8 and 9 show what that costs.

3. In R

```
M3 <- cbind(a = c(1, 2, 3), b = c(2, 4, 6)) # b = 2a: dependent
qr(M3)$rank
qr(cbind(a = c(1, 2, 3), b = c(2, 4, 7)))$rank # change one entry
```

```
[1] 1
[1] 2
```

- `qr(M)` factorises a matrix and returns a *list*; `$rank` is the one element we read. It is R's rank test (1 for the dependent pair, 2 once one entry changes).

? Predict 5 — answer before you read on

Add a fifth column to `X`: `dep = pop0014 + pop65`, the share of the population that is under 15 or over 65 — a real quantity, and exactly a sum of two columns. What is the rank of the 175×5 matrix?

```
X5 <- cbind(X, dep = X[, "pop0014"] + X[, "pop65"]) # 5th col: a SUM
c(rank_X = qr(X)$rank, rank_Xc = qr(Xc)$rank, rank_X5 = qr(X5)$rank)
S5 <- cov(X5)
round(S5, 2)
```

```
rank_X rank_Xc rank_X5
      4      4      4
      pop0014 pop65 life_exp gdp_k dep
pop0014  110.22 -61.31 -70.56 -160.73 48.91
pop65    -61.31  51.03  38.68  119.28 -10.28
life_exp -70.56  38.68  67.28  130.24 -31.88
gdp_k    -160.73 119.28 130.24 801.64 -41.45
dep       48.91 -10.28 -31.88 -41.45 38.63
```

- `cbind(X, dep = ...)` glues a fifth column on. Its rank is **4**, **not 5**: $\mathbf{x}_1 + \mathbf{x}_2 - \mathbf{x}_5 = \mathbf{0}$ exactly. The centred matrix `Xc` also has rank 4 — here $p = 4 < n - 1$, so centring cost nothing.
- `S5` is a 5×5 table that prints without complaint. Nothing about it *looks* broken — the redundancy is invisible to the eye. It becomes visible in Sections 8 and 9.

```
X3 <- X[c("Qatar", "Kuwait", "Bahrain"), ] # n = 3 countries, p = 4
qr(scale(X3, scale = FALSE))$rank         # centred: at most n - 1
qr(cov(X3))$rank                           # so S is 4 x 4 of rank 2
```

```
[1] 2
[1] 2
```

- Three countries, four columns: after centring the rank is $2 = n - 1$, and the 4×4 covariance matrix has rank 2. No redundancy among the variables — just too few rows.

7 Trace

In this section

The question. *Is there one number that says how much the whole table varies?*

What we do about it. Add the diagonal of `S`. Then show what that number cannot see.

1. Definition

Definition 7.1: Trace (J&W Definition 2A.28, Result 2A.12)

- The **trace** of a square matrix is the sum of its diagonal entries:

$$\text{tr}(\mathbf{A}) = \sum_{i=1}^k a_{ii}.$$

Definition 7.1: Trace (J&W Definition 2A.28, Result 2A.12) (continued)

- It is linear, and a product may be turned around inside it:

$$\text{tr}(c\mathbf{A}) = c \text{tr}(\mathbf{A}), \quad \text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B}), \quad \text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA}).$$

- **Tiny example.** $\text{tr} \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix} = 2 = \text{tr} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ — the off-diagonal does not enter.

2. In statistics

Fact 7.1: $\text{tr}(\mathbf{S})$ is the total sample variance

- $\text{tr}(\mathbf{S}) = s_{11} + \cdots + s_{pp}$: the variances added up. J&W call it the **total sample variance** (3-23).
- It **ignores the correlations**: the tiny example is two covariance matrices with the same total variance, one with $r = 0$ and one with $r = 0.9$.
- It is in *mixed units* when the columns are: our $\text{tr}(\mathbf{S})$ adds percent-squared to years-squared to thousand-dollars-squared. $\text{tr}(\mathbf{R}) = p$ always.

3. In R

```
c(trace = sum(diag(S)), total_variance = sum(apply(X, 2, var)))
round(diag(S) / sum(diag(S)), 3) # each column's share
```

```
      trace total_variance
      1030      1030
pop0014  pop65 life_exp  gdp_k
  0.107   0.050   0.065   0.778
```

- $\text{sum}(\text{diag}(\mathbf{S}))$ is the trace; $\text{apply}(\mathbf{X}, 2, \text{var})$ (Lecture 5: 2 = columns) computes the four variances separately and their sum agrees: 1030.
- **Income alone is 78% of the total variance** — because it is in thousand dollars. In dollars (Section 4) the share would be 99.99%. A total variance follows the loudest unit.

```
S_a <- matrix(c(1, 0, 0, 1), 2); S_b <- matrix(c(1, 0.9, 0.9, 1), 2)
c(trace_a = sum(diag(S_a)), trace_b = sum(diag(S_b)))
c(cor_a = cov2cor(S_a)[1, 2], cor_b = cov2cor(S_b)[1, 2])
all.equal(sum(diag(A %*% B)), sum(diag(B %*% A))) # tr(AB) = tr(BA)
```

```
trace_a trace_b
      2      2
cor_a cor_b
  0.0  0.9
[1] TRUE
```

- $\text{cov2cor}()$ converts a covariance matrix into its correlation matrix (the sandwich of Section 4 in one call). Same trace, correlation 0 against 0.9: the trace cannot tell the two apart.
- $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$ checked on the 2×2 and the 3×3 products of Section 2: different sizes, same trace.

8 Determinant

In this section

The question. What is the “generalized sample variance” — a name used once and put off — and what does it see that the trace does not?

What we do about it. Define the determinant, compute $|\mathbf{S}|$ and $|\mathbf{R}|$, and watch $|\mathbf{R}|$ fall from 1 to 0 as correlations grow — then collapse on the five-column table of Section 6.

1. Definition

Definition 8.1: Determinant (J&W Definition 2A.24, Result 2A.11)

- For a 2×2 matrix the **determinant** is

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

For $k \times k$ it is defined by expanding along a row (J&W 2A.24); R’s `det()` computes it.

- Properties we use:

$$|\mathbf{A}^\top| = |\mathbf{A}|, \quad |\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|, \quad |c\mathbf{A}| = c^k|\mathbf{A}|, \quad |\mathbf{A}| = 0 \iff \mathbf{A} \text{ is singular.}$$

- Tiny example** (J&W Example 2.8): $\begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix} = 3 \cdot 1 - 2 \cdot 4 = -5.$

2. In statistics

Fact 8.1: $|\mathbf{S}|$ is the generalized sample variance; $|\mathbf{R}| \in [0, 1]$

- $|\mathbf{S}|$ is J&W’s **generalized sample variance** (§3.4): one number for the spread of the whole cloud, *including* how the columns move together.
- From the sandwich of Section 4 and $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$:

$$|\mathbf{S}| = |\mathbf{D}^{1/2}| |\mathbf{R}| |\mathbf{D}^{1/2}| = (s_{11}s_{22} \cdots s_{pp}) |\mathbf{R}|, \quad \text{so} \quad |\mathbf{R}| = \frac{|\mathbf{S}|}{s_{11} \cdots s_{pp}} \quad (\text{J\&W 3-21}).$$

- $|\mathbf{R}|$ is **unit-free** and lies in $[0, 1]$: it is 1 when every correlation is 0, and it falls toward 0 as the columns become predictable from each other. It is 0 exactly when $\mathbf{S}$ is singular (Section 6).
- For $p = 2$: $|\mathbf{R}| = 1 - r^2$.

3. In R

```
M2 <- matrix(c(3, 4, 2, 1), 2) # J&W Example 2.8
M2
det(M2) # 3 * 1 - 2 * 4
c(det_t = det(t(M2)), det_2M = det(2 * M2), det_MM = det(M2 %*% M2))
```

```
      [,1] [,2]
[1,]    3    2
[2,]    4    1
[1] -5
det_t det_2M det_MM
-5    -20    25
```

- `det()` takes one square matrix. The checks: the transpose keeps -5 ; $2 * M2$ gives $2^2 \times (-5) = -20$; the product gives $(-5)^2 = 25$.

```
c(det_S = det(S), det_R = det(R_hand), ratio = det(S) / prod(diag(S)))
```

```
      det_S      det_R      ratio
1.971e+07 6.497e-02 6.497e-02
```

- $|S| \approx 1.97 \times 10^7$: a large number in a unit with no name ($\text{percent}^2 \times \text{percent}^2 \times \text{years}^2 \times \text{thousand dollars}^2$). `prod(diag(S))` divides the units out and leaves $|R| = 0.065$ — the same number `det(R_hand)` returns.
- 0.065 on a scale from 0 to 1: the four columns are far from independent. Section 5 already showed two of them at 144.8° .

For a 2×2 correlation matrix, $|R| = 1 - r^2$. Four values of r :

```
for (r in c(0, 0.5, 0.9, 0.99))
  cat("r =", r, " |R| =", det(matrix(c(1, r, r, 1), 2)), "\n")
```

```
r = 0      |R| = 1
r = 0.5    |R| = 0.75
r = 0.9    |R| = 0.19
r = 0.99   |R| = 0.0199
```

- `for (r in ...)` runs the line once per value; `cat()` prints the pieces on one line. The ladder is $1 - r^2$: 1, 0.75, 0.19, 0.0199. The last step from 0.9 to 0.99 costs a factor of ten.

? Predict 6 — answer before you read on

`det(S5)`, the five-column table of Section 6 with its exact sum column. Exactly 0?

```
det(S5)                                     # the 5-column S of Section 6
```

```
[1] 1.89e-06
```

- **Not exactly 0:** 1.9×10^{-6} . The true value is 0 — the rank is 4 — but the computer works to about 16 digits, and S_5 has entries in the hundreds, so “zero” comes out as rounding dust.
- Read this rule off the screen: **`det()` is not a singularity test.** The test is `qr()$rank`, or the refusal of Section 9.

9 Inverse

In this section

The question. *Lecture 5’s Mahalanobis distance had S^{-1} in the middle and we called it “the matrix that undoes S ”. What is it, when does it exist, and what did `solve()` refuse to do on Worksheet 5?*

What we do about it. Define the inverse, compute S^{-1} , check it, write the Mahalanobis distance properly and recompute one of Lecture 5’s numbers by hand. Then ask for the inverse of the singular table and read the error.

1. Definition

Definition 9.1: Inverse (J&W 2-12, Results 2A.9, 2A.10)

- The **inverse** of a square $\mathbf{A}$ is the matrix $\mathbf{A}^{-1}$ with

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}.$$

- It exists exactly when the columns of $\mathbf{A}$ are linearly independent — the three statements are equivalent:

$$\mathbf{A}^{-1} \text{ exists} \iff |\mathbf{A}| \neq 0 \iff \text{rank}(\mathbf{A}) = k.$$

- Two properties:

$$(\mathbf{A}\mathbf{B})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}, \quad \text{diag}(d_1, \dots, d_k)^{-1} = \text{diag}(1/d_1, \dots, 1/d_k).$$

- Tiny example** (J&W Example 2.8):

$$\begin{pmatrix} 3 & 2 \\ 4 & 1 \end{pmatrix}^{-1} = \frac{1}{-5} \begin{pmatrix} 1 & -2 \\ -4 & 3 \end{pmatrix} = \begin{pmatrix} -0.2 & 0.4 \\ 0.8 & -0.6 \end{pmatrix}, \quad \text{check: } \begin{pmatrix} -0.2 & 0.4 \\ 0.8 & -0.6 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 4 & 1 \end{pmatrix} = \mathbf{I}.$$

2. In statistics

Fact 9.1: The Mahalanobis distance, written properly

- Lecture 5's squared statistical distance of a country $\mathbf{x}$ from the centre $\bar{\mathbf{x}}$ is

$$d^2(\mathbf{x}, \bar{\mathbf{x}}) = \underbrace{(\mathbf{x} - \bar{\mathbf{x}})^{\top}}_{(1 \times p)} \underbrace{\mathbf{S}^{-1}}_{(p \times p)} \underbrace{(\mathbf{x} - \bar{\mathbf{x}})}_{(p \times 1)} \quad \text{— one number.}$$

- It exists only when $\mathbf{S}^{-1}$ does: full rank, $|\mathbf{S}| \neq 0$. A redundant column, or $n \leq p$, and there is no distance to compute — that is the error Lecture 5 warned about.
- When $\mathbf{S}$ is diagonal, $\mathbf{S}^{-1} = \text{diag}(1/s_{jj})$ and $d^2 = \sum_j (x_j - \bar{x}_j)^2 / s_{jj}$: the squared *standardised Euclidean* distance of Lecture 5. The off-diagonal of $\mathbf{S}^{-1}$ is where the correlations act.

3. In R

```
solve(M2)           # J&W Example 2.8: the inverse
zapsmall(solve(M2) %*% M2) # A^{-1} A = I (rounding dust set to 0)
solve(diag(c(2, 4, 5))) # a diagonal matrix: invert the diagonal
```

```
      [,1] [,2]
[1,] -0.2  0.4
[2,]  0.8 -0.6
      [,1] [,2]
[1,]  1    0
[2,]  0    1
      [,1] [,2] [,3]
[1,]  0.5 0.00  0.0
[2,]  0.0 0.25  0.0
[3,]  0.0 0.00  0.2
```

- `solve(A)` with one argument returns $\mathbf{A}^{-1}$ (with two, `solve(A, b)`, it solves $\mathbf{A}\mathbf{x} = \mathbf{b}$ instead). `zapsmall()` sets entries that are zero up to rounding dust to exactly 0, so the identity prints as an identity.
- A diagonal matrix inverts entry by entry: 1/2, 1/4, 1/5.

```
S_inv <- solve(S)
round(S_inv, 4)
zapsmall(S_inv %*% S)           #  $S^{-1}S = I$ , tiny rounding set to 0
```

	pop0014	pop65	life_exp	gdp_k
pop0014	0.0473	0.0353	0.0308	-0.0008
pop65	0.0353	0.0660	0.0064	-0.0038
life_exp	0.0308	0.0064	0.0487	-0.0027
gdp_k	-0.0008	-0.0038	-0.0027	0.0021

	pop0014	pop65	life_exp	gdp_k
pop0014	1	0	0	0
pop65	0	1	0	0
life_exp	0	0	1	0
gdp_k	0	0	0	1

- S^{-1} exists (rank 4, $|S| \neq 0$) and $S^{-1}S$ prints as I_4 . Always run this check when a computer hands you an inverse (J&W, p.59).
- Its entries are not “one over the variances”: the pop0014 entry is 0.047, while $1/110.2 = 0.009$. The correlations have moved everything.

? Predict 7 — answer before you read on

Qatar's d^2 from the mean of the four columns, computed with the full S^{-1} — and computed with only the diagonal of S . Which is larger?

```
g <- X["Qatar", ] - colMeans(X)           # Qatar minus the mean
c(by_hand = t(g) %*% S_inv %*% g,
  mahalanobis = mahalanobis(X["Qatar", ], colMeans(X), S))
c(diagonal_S = mahalanobis(X["Qatar", ], colMeans(X), diag(diag(S))),
  std_euclid_sq = sum((g / sqrt(diag(S)))^2))
```

by_hand	mahalanobis
26.57	26.57
diagonal_S	std_euclid_sq
9.986	9.986

- $t(g) \% \% S_inv \% \% g$ is the formula typed as written; `mahalanobis(x, center, cov)` — the unit, the point measured from, the covariance matrix (*not* its inverse) — returns the same 26.57.
- With the diagonal only, 9.99. The full S^{-1} more than doubles Qatar's distance: Qatar is rich *and* young, and the columns say those two do not go together. That is what the off-diagonal entries of S^{-1} measure.

```
wh <- read.csv("data/worldhealth2022.csv", row.names = 1)   # Lecture 5
X8 <- as.matrix(wh[, 4:11]); rownames(X8) <- wh$country
d <- X8["Qatar", ] - X8["United Arab Emirates", ]          # the gap vector
c(by_hand = t(d) %*% solve(cov(X8)) %*% d,
  L5 = mahalanobis(X8["Qatar", ], X8["United Arab Emirates", ],
    cov(X8)))
```

by_hand	L5
4.173	4.173

- Lecture 5, Question 3.2, on its eight columns: Qatar to the United Arab Emirates, 4.172691. Same number, now from the formula with `solve()` in the middle. The line Lecture 5 could only run is explained.

```
solve(S5)
```

```
# the singular S of Section 6
```

```
Error in solve.default(S5) : system is computationally singular: reciprocal  
condition number = 2.5506e-17
```

- Read the message: *computationally singular* — R found the rank deficiency of Section 6; *reciprocal condition number* 2.6×10^{-17} — a number of order machine precision, R’s way of saying “the determinant is zero up to rounding”.
- This is the refusal of Lecture 5’s Home Problem 2 and of Worksheet 5. The cause is always the same: a column that is a combination of the others, or fewer rows than columns.

10 Four named matrices

In this section

The question. *Four kinds of square matrix keep appearing. What are they, and what does the last one — the rotation — do to a data cloud?*

What we do about it. Name them, say which of today’s objects belongs to which, and rotate two columns of the data by 30° to see what changes and what does not. **What the section leaves behind:** the question Lecture 7 answers.

1. Definition

Definition 10.1: Symmetric, diagonal, identity, orthogonal (J&W 2-11, 2-13)

- **Symmetric:** $\mathbf{A}^\top = \mathbf{A}$, i.e. $a_{ij} = a_{ji}$.
- **Diagonal:** zeros off the diagonal (Section 4).
- **Identity $\mathbf{I}$:** the diagonal matrix of ones; $\mathbf{IA} = \mathbf{AI} = \mathbf{A}$.
- **Orthogonal $\mathbf{Q}$:** its transpose is its inverse,

$$\mathbf{Q}^\top \mathbf{Q} = \mathbf{Q} \mathbf{Q}^\top = \mathbf{I},$$

so every column of $\mathbf{Q}$ has length 1 and any two columns are perpendicular. The rotation by an angle θ is the standard example:

$$\mathbf{Q} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad \mathbf{Q}^\top \mathbf{Q} = \begin{pmatrix} \cos^2 \theta + \sin^2 \theta & 0 \\ 0 & \cos^2 \theta + \sin^2 \theta \end{pmatrix} = \mathbf{I}.$$

2. In statistics

Fact 10.1: Which is which, and what a rotation leaves alone

- $\mathbf{S}$ and $\mathbf{R}$ are **symmetric** because $s_{jk} = s_{kj}$: the covariance of j with k is the covariance of k with j . $\mathbf{H}$ is symmetric too. $\mathbf{X}_c$ is not even square.
- $\mathbf{D}$ is **diagonal**; $\mathbf{I}$ is the covariance matrix of standardised uncorrelated columns (Lecture 5’s “when Euclidean equals Mahalanobis”).
- If the data are turned, $\mathbf{Y} = \mathbf{X}_c \mathbf{Q}$ with $\mathbf{Q}$ orthogonal, then the covariance matrix turns with them,

$$\mathbf{S}_\mathbf{Y} = \frac{1}{n-1} \mathbf{Y}^\top \mathbf{Y} = \mathbf{Q}^\top \mathbf{S} \mathbf{Q},$$

and **every distance between two countries, the trace and the determinant are unchanged:** $\text{tr}(\mathbf{Q}^\top \mathbf{S} \mathbf{Q}) = \text{tr}(\mathbf{S} \mathbf{Q} \mathbf{Q}^\top) = \text{tr}(\mathbf{S})$ and $|\mathbf{Q}^\top \mathbf{S} \mathbf{Q}| = |\mathbf{Q}|^2 |\mathbf{S}| = |\mathbf{S}|$.

Fact 10.1: Which is which, and what a rotation leaves alone (continued)

- The individual entries of **S** *do* change. Lecture 7 asks which rotation makes them all zero off the diagonal.

3. In R

```
c(S = isSymmetric(S), R = isSymmetric(R_hand), Xc = isSymmetric(Xc))
th <- 30 * pi / 180 # a rotation by 30 degrees
Q <- matrix(c(cos(th), sin(th), -sin(th), cos(th)), 2)
round(Q, 3)
zapsmall(t(Q) %*% Q) # Q'Q = I: orthogonal
```

```
      S      R      Xc
TRUE  TRUE FALSE
      [,1] [,2]
[1,] 0.866 -0.500
[2,] 0.500  0.866
      [,1] [,2]
[1,] 1      0
[2,] 0      1
```

- `isSymmetric()` tests $\mathbf{A}^\top = \mathbf{A}$; **S** and **R** pass, the 175×4 **Xc** cannot.
- **Q** is the 30° rotation ($\cos 30^\circ = 0.866$, $\sin 30^\circ = 0.5$), and `t(Q) %*% Q` is the identity.

? Predict 8 — answer before you read on

Rotate the two centred columns `pop65` and `life_exp` by 30° . Does the covariance matrix change? Does its trace? Do the distances between countries?

```
P <- Xc[, c("pop65", "life_exp")] # two centred columns
Y <- P %*% Q # every country, turned
c(trace_before = sum(diag(cov(P))), trace_after = sum(diag(cov(Y))))
c(det_before = det(cov(P)), det_after = det(cov(Y)))
max(abs(dist(P) - dist(Y))) # every distance: unchanged
round(cov(Y), 3) # but the covariance moved
```

```
trace_before trace_after
      118.3      118.3
det_before det_after
      1937      1937
[1] 1.421e-14
      [,1] [,2]
[1,] 88.59 26.38
[2,] 26.38 29.72
```

- `P %*% Q` applies the rotation to all 175 rows at once. Trace 118.3 before and after; determinant 1937 before and after; the largest change in any of the $175 \times 174/2$ distances is 10^{-14} — rounding dust.
- Yet `cov(Y)` is a new matrix: variances 88.6 and 29.7 instead of 51.0 and 67.3, covariance 26.4 instead of 38.7. **The cloud did not change; only the axes did.**
- Lecture 7 asks: *which* rotation makes the off-diagonal entry exactly 0 — and what the two variances are then.

11 Warm-up

In this section

The question. *Did any of that stick?*

What we do about it. Five lines of R on a 2×3 matrix of the numbers 1 to 6. Predict each output, then run it. The answers follow — try the five first.

Warm-up — five lines, predict each output before you press Enter

1. `W1 <- matrix(1:6, nrow = 2); dim(W1)`
2. `dim(t(W1))`
3. `W1 %*% t(W1)` — what size, and what is its trace?
4. `t(W1) %*% W1` — what size, and what is its trace?
5. `diag(2) %*% W1`

```
W1 <- matrix(1:6, nrow = 2)
dim(W1)
dim(t(W1))
W1 %*% t(W1)
t(W1) %*% W1
diag(2) %*% W1
```

```
[1] 2 3
[1] 3 2
      [,1] [,2]
[1,]    35   44
[2,]    44   56
      [,1] [,2] [,3]
[1,]     5   11  17
[2,]    11   25  39
[3,]    17   39  61
      [,1] [,2] [,3]
[1,]     1    3    5
[2,]     2    4    6
```

- 2×3 , then 3×2 . `W1 %*% t(W1)` is 2×2 with trace $35 + 56 = 91$; `t(W1) %*% W1` is 3×3 with trace $5 + 25 + 61 = 91$ — $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$ (Section 7). The identity changes nothing.

12 Problems to try at home — all in R

In this section

The question. *Does everything from today survive on other columns of the same file?*

What we do about it. Five problems on four columns of `whwide2022` not used above. Each asks you to run something and write **two sentences** about what came back. Short answers at the end of the section; full solutions in the practice session.

The columns: `birth_rate` and `death_rate` (per 1000 people), `mobile` (subscriptions per 100 people), `internet_pct` (%). Build them as in Section 1:

```
U <- as.matrix(w[, c("birth_rate", "death_rate", "mobile",
                    "internet_pct")])
rownames(U) <- w$country
```

Home Problem 1. S by hand. Build the 175×175 centring matrix, centre U , and compute $S = U_c^\top U_c / (n - 1)$. Check it against `cov(U)` with `all.equal()`, then report `tr(S)` and each column's share of it. Two sentences: *which column dominates the total variance, and is that a fact about countries or about units?*

Home Problem 2. An angle. Compute the cosine of the angle between the centred `birth_rate` and `death_rate` columns from lengths and an inner product, compare it with `cor(U)`, and convert it to degrees. Two sentences: *how far from a right angle is it, and what does that say about the two rates across countries?*

Home Problem 3. An exact redundancy. Build $V2$ from the two columns `urban_pct` and `rural_pct`, as U was built. Print `range(rowSums(V2))`, then `qr(V2)$rank`, the rank of the centred $V2$, `det(cov(V2))` and finally `solve(cov(V2))`. Two sentences: *why does the rank drop by one more after centring, and what does R say when asked for the inverse?*

Home Problem 4. A redundancy that nearly holds. Build $V3$ from `dep_old`, `dep_young` and `dep_ratio`, in that order. The codebook says the third column is the sum of the first two. Print `range(V3[, 3] - V3[, 1] - V3[, 2])`, `qr(V3)$rank`, `det(cov(V3))` and `round(solve(cov(V3)))`. Two sentences: *did R refuse this time — and would you trust the inverse it returned?*

Home Problem 5. Two determinants and one distance. On U : compute $|R|$ for all four columns and for the pair `mobile`, `internet_pct` alone. Then compute Qatar's d^2 from the mean vector with `t(g) %*% solve(cov(U)) %*% g` and with `mahalanobis()`. Two sentences: *why is the four-column $|R|$ smaller than the two-column one, and do the two d^2 agree?*

*Short answers. 1) TRUE; $\text{tr}(S) = 2019$; `mobile` carries 61.8% and `internet_pct` 32.1% — mostly units: subscriptions per 100 vary over a range of about 150, the two rates over about 30. 2) $\cos \theta = r = -0.036$, $\theta = 92.0^\circ$ — two degrees from perpendicular: across countries the birth rate tells you almost nothing about the death rate. 3) Every row sums to exactly 100; rank 2 raw (the columns are not multiples of each other) but 1 after centring (the centred columns are exact negatives); `det` prints -5×10^{-11} — rounding dust, and even the sign is noise; `solve()` stops with system is computationally singular. 4) The gap is within ± 0.001 (rounding to three decimals), so the rank is 3, $|S| = 0.0078$, and **R does not refuse** — it returns an inverse with entries around ± 4 million. An identity that nearly holds is worse than one that holds exactly: the inverse exists and is meaningless. 5) $|R_4| = 0.160$, $|R_2| = 0.698$; adding columns can only add dependence, so $|R|$ falls. Qatar: 4.328 both ways.*

13 FAQ

In this section

The question. *The questions this lecture usually leaves behind — answered in two lines each.*

What we do about it. Read these after the home problems; each answer points back to the section that earned it.

FAQ — Lecture 6

1. Is $X^\top$ the same as $t(X)$? And why does J&W write X' ?

Yes — three notations for one operation: swap rows and columns. This course writes $^\top$; the book writes a prime (Section 1).

2. How do I know whether `A %*% B` will run?

Write the dimensions under each factor. The inner pair must match; the outer pair is the size of the answer. If the inner pair does not match, R says *non-conformable arguments* (Section 2).

FAQ (continued)

3. **Why $n - 1$ in $\mathbf{S} = \mathbf{X}_c^\top \mathbf{X}_c / (n - 1)$?**

The same convention as `var()` and `sd()` since Lecture 2; J&W's $\mathbf{S}_n$ divides by n . Every ratio in this lecture — $\mathbf{R}$, $|\mathbf{R}|$, d^2 — is the same under either convention (Section 3).

4. **Is `cov(scale(X))` really `cor(X)`?**

Yes, to the last digit: standardised columns have variance 1, so the sandwich $\mathbf{D}^{-1/2} \mathbf{S} \mathbf{D}^{-1/2}$ has nothing left to divide by (Section 4).

5. **A correlation of -0.8 is an angle of 145° . In what space?**

In the space of 175-entry columns: each variable is one arrow with 175 coordinates, and the angle between two arrows is what `cor()` reports through its cosine (Section 5).

6. **`det(S)` printed 10^{-6} , not 0. Is $\mathbf{S}$ singular or not?**

Singular. `det()` works to about 16 digits and a true zero surfaces as dust. Use `qr(S)$rank`, or let `solve()` refuse (Sections 8, 9).

7. **Which is the right “size of the spread”: `tr(S)` or `|S|`?**

They answer different questions. The trace adds the variances and cannot see correlation; the determinant shrinks as the columns become predictable from each other. Compare units-free versions, $\text{tr}(\mathbf{R}) = p$ and $|\mathbf{R}| \in [0, 1]$ (Sections 7, 8).

8. **`mahalanobis(x, center, cov)`: do I pass $\mathbf{S}$ or $\mathbf{S}^{-1}$?**

$\mathbf{S}$. The function inverts it internally; passing `solve(S)` gives a wrong number silently (Section 9).

9. **If a rotation changes every entry of $\mathbf{S}$, why say the data “did not change”?**

Because every distance between two countries, the trace and the determinant are identical after the rotation. The countries are where they were; the axes turned (Section 10).

Wrap-up

Summary of Lecture 6

- **The table is a matrix $\mathbf{X}$** , 175×4 ; a variable is a column, a country is a row; $\bar{\mathbf{x}} = \mathbf{X}^\top \mathbf{1}/n$.
- **Products need matching inner dimensions**; the inner product of two columns is a sum of cross-products, the outer product a table.
- **The spine: $\mathbf{X}_c = \mathbf{H}\mathbf{X}$ and $\mathbf{S} = \mathbf{X}_c^\top \mathbf{X}_c/(n-1)$** , matching `cov()` exactly.
- **$\mathbf{R} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2}$** : a unit change is a diagonal sandwich; **$\mathbf{S}$ moves, $\mathbf{R}$ does not.**
- **Correlation is a cosine** — so $|r| \leq 1$ is Cauchy–Schwarz. `pop0014` and `pop65` sit at 144.8° .
- **Rank**: a sum column costs one; $n \leq p$ costs more. The 5-column table has rank 4.
- **Trace** = total variance, blind to correlation (78% of ours is income). **Determinant** = generalized variance; $|\mathbf{R}| = 0.065$; `det()` prints dust, not zero, on a singular matrix.
- **Inverse**: exists iff full rank iff $|\mathbf{S}| \neq 0$; $d^2 = (\mathbf{x} - \bar{\mathbf{x}})^\top \mathbf{S}^{-1}(\mathbf{x} - \bar{\mathbf{x}})$, Qatar 26.57 (diagonal only: 9.99); Lecture 5's 4.172691 recovered. `solve()` refuses the singular table.
- **Orthogonal $\mathbf{Q}$ = a rotation**: distances, trace, determinant unchanged; the entries of **$\mathbf{S}$** change.
- **The R to remember**: `%*`, `t()`, `crossprod()`, `diag()` (three jobs), `qr()$rank`, `det()`, `solve()`, `zapsmall()`, `all.equal()`.

Next lecture (Tue, Sep 15): *Lecture 7 — which rotation makes the covariance matrix diagonal, and what the variances are then.*